

z-test ✓ → t-test ✓

problem statement

Do you know the population variance (σ)

yes

no

t-test

n > 30

yes

no

z-test

t-test

Sample Size
n > 30

σ

σ ✓ n > 30

↓

z-test

x

S ✓ n > 30

false

↓

t-test

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}} \quad \leftarrow x$$

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$